

A Multi-Layer Dynamic Theory of Motivation: Integrating Psychological Needs, Valence–Arousal Representation, and Behavioral Dynamics

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1 Introduction

In contemporary educational and psychological research, motivation is regarded as a key psychological mechanism that influences the initiation of behavior, sustained engagement, and learning performance. In recent years, motivation research has gradually shifted from a static trait-oriented perspective to a dynamic and context-integrated approach, emphasizing the variability and plasticity of motivation in the continuous interaction between individuals and their environments (Pintrich, 2003; Eccles & Wigfield, 2002). Relevant studies indicate that an individual’s level of motivation is influenced not only by their perceptions of task value and expectancy for success, but also by task costs, social contextual support, and immediate emotional experiences (Wigfield & Eccles, 2000; Barron & Hulleman, 2015). Meanwhile, research based on Self-Determination Theory further suggests that when individuals experience autonomy, competence, and relatedness during activities, their intrinsic motivation is more likely to be activated and sustained (Deci & Ryan, 2000; Ryan & Deci, 2017).

However, despite substantial progress in explaining the mechanisms underlying the formation of individual motivation, existing theories still exhibit several limitations. First, current models still tend to rely on static or structural explanatory frameworks, limiting their ability to capture the dynamic and temporally evolving nature of motivation in real behavioral processes (Pintrich, 2003; Boekaerts, 2011; Murayama et al., 2017). Second, although a range of variables such as expectancy, value, cost, and autonomy have been incorporated, a coherent account of how these factors are integrated and interact with one another remains underdeveloped (Barron & Hulleman, 2015; Eccles & Wigfield, 2020). Finally, although the importance of situational factors and individual differences is widely acknowledged, the mechanisms through which contextual conditions systematically shape motivational structures and behavioral outcomes remain poorly specified (Mischel & Shoda, 1995; Ryan & Deci, 2017; Reeve, 2012).

Therefore, this paper aims to extend and integrate the aforementioned limitations by proposing a multi-layer theoretical reconstruction of motivation.

First, within an expanded framework of Self-Determination Theory, basic psychological needs are reconceptualized into four distinct categories—instinctual, experiential, state, and interactive needs—in order to capture a more fine-grained hierarchical and dynamic structure of motivational sources (Deci & Ryan, 2000; Ryan & Deci, 2017; Reeve, 2012).

Second, drawing on the valence–arousal framework (Russell, 1980; Barrett & Russell, 2009), this study treats valence–arousal as a representational system for motivational experience rather than merely an emotional descriptor. Specifically, valence is used to encode the positive–negative quality of both experienced and anticipated outcomes, while arousal reflects the required intensity of effort investment as well as its temporal persistence. In this sense, motivational states can be characterized in terms of both evaluative direction (valence) and resource-related demand (arousal) within a unified representational space.

Finally, these valence–arousal representations are embedded into a temporal dynamic process to capture the continuous evolution of motivational states over time. The resulting trajectories are further linked to behavioral decision-making processes, thereby specifying how changes in subjective motivational experience are transformed into observable behavioral choices (Mischel & Shoda, 1995; Murayama et al., 2017; Smith & Conrey, 2007).